Rockchip RealTime Linux Performance Test Report

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Preface

With the increasing demand for real-time performance of products, the real-time performance of standard Linux can no longer meet the needs of many products. It is necessary to optimize standard Linux to improve real-time performance, such as: PREEMPT_RT patches, Xenomai real-time system.

Overview

This document provides real-time performance test data of Rockchip chips under different RT Linux implementations.

Product Version

Chip Name	Kernel Version
RK3568	kernel-5.10
RK3562	kernel-5.10
RK3588	kernel-5.10
RK3576	kernel-6.1
RK3506	kernel-6.1
RV1126B	kernel-6.1
RK3572	kernel-6.12

Target Audience

This document (this guide) is mainly applicable to the following engineers:

Technical Support Engineers

Software Development Engineers

Hardware Development Engineers

Revision Record

Version Number	Author	Modification Date	Modification Description
V1.0.0	Chen Liang	2023-08-14	Initial version
V1.0.1	ZhiZhan Chen	2023-10-07	Modified RK3568 related description
V1.0.2	Caesar Wang	2024-06-20	Added RK3576 support Modified some section contents
V1.1.0	Chen Liang	2024-06-20	Added RK3506 support
V1.2.0	Linjianhua	2025-04-10	Update RK3568 Preempt-RT data
V1.3.0	Linjianhua	2025-06-10	Added RV1126B support
V1.4.0	Chen Liang	2026-06-12	Added RK3572 support Update RK3576 Test Data

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1. Introduction to Real-Time Linux

The standard Linux system aims to construct a complete and stable open-source operating system, striving to reduce the average system response time and enhance throughput. It focuses on the overall functional requirements of the operating system to achieve better average performance. Therefore, standard Linux does not offer hard real-time capabilities.

To address the issue of Linux's lack of hard real-time features, several hard real-time solutions based on Linux have emerged, which can be categorized into two types:

- Direct modification of the Linux kernel source code: Making subtle changes to the Linux kernel code without extensive alterations, and transforming Linux into a fully preemptive real-time system by directly modifying the kernel source code in compliance with the GPL license. The disadvantage is that by modifying the Linux kernel, it is difficult to ensure that the execution of real-time processes will not be disturbed by unpredictable activities of non-real-time processes. The representative of this method is RT-patch (Real Preemption Patch).
- Dual-kernel approach: Adding a real-time kernel that coexists with the Linux kernel in the kernel space, and scheduling the standard Linux kernel as a regular process on the real-time kernel. The advantage is that it can achieve hard real-time performance and conveniently implement a new scheduling strategy. Common dual-kernel methods include RT-Linux, RTAI (Real-Time Application Interface), and Xenomai, which offer good real-time performance.

2. Real-time Performance Testing

2.1 RK3568

Configuration	Description
CPU	4xA55, 1314 MHz (PVTPLL@0.9V@len=0x43)
CACHE	L1-32K, L2-0K, L3-512K
DDR	DDR4, 1560M
OS	BUILDROOT, kernel-5.10
BL31	rk3568_bl31_rt_v1.03.elf (optimize RT latency)

2.1.1 No-Load Test

2.1.1.1 PREEMPT_RT Patch

```
rk3568_t:/ # cyclictest -c 0 -m -a -t 4 -p 99
policy: fifo: loadavg: 0.59 0.56 0.98 2/245 2084

T: 0 ( 1976) P:99 I:1000 C:84680446 Min:      0 Act:    0 Avg:    0 Max:      21
T: 1 ( 1977) P:99 I:1500 C:56453626 Min:      0 Act:    0 Avg:    0 Max:      17
T: 2 ( 1978) P:99 I:2000 C:42340216 Min:      0 Act:    0 Avg:    0 Max:      22
T: 3 ( 1979) P:99 I:2500 C:33872169 Min:      0 Act:    1 Avg:    1 Max:      18
(Testing for 24 hour)
```

2.1.1.2 Xenomai Cobalt Mode

```
rk3568_t:/ # echo 0 > /proc/xenomai/latency
rk3568_t:/ # /usr/demo/cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0 microseconds
policy: fifo: loadavg: 0.29 0.18 0.12 1/1107 4216

T: 0 ( 4021) P:99 I:1000 C:12501264 Min:      1 Act:    8 Avg:    7 Max:      68
T: 1 ( 4022) P:99 I:1500 C:8334176 Min:      2 Act:    8 Avg:    7 Max:      70
T: 2 ( 4023) P:99 I:2000 C:6250631 Min:      2 Act:    9 Avg:    8 Max:      66
T: 3 ( 4024) P:99 I:2500 C:5000505 Min:      2 Act:    6 Avg:    7 Max:      58
(Testing for 3 hours)
```

2.1.2 Pressure Test 1

`stress-ng -c 4 --io 2 --vm 1 --vm-bytes 4M --timeout 1000000s`

2.1.2.1 PREEMPT_RT Patch

```
rk3568_t:/ # cyclictest -c 0 -m -a -t 4 -p 99
policy: fifo: loadavg: 10.72 10.57 10.54 10/264 3892

T: 0 ( 2023) P:99 I:1000 C:76581100 Min:      0 Act:    2 Avg:    3 Max:      74
T: 1 ( 2024) P:99 I:1500 C:51054043 Min:      0 Act:    2 Avg:    3 Max:      64
T: 2 ( 2025) P:99 I:2000 C:38290514 Min:      0 Act:    4 Avg:    4 Max:      44
T: 3 ( 2026) P:99 I:2500 C:30632399 Min:      0 Act:    2 Avg:    4 Max:      61
(Testing for 24 hours)
```

2.1.2.2 Xenomai Cobalt Mode

```
rk3568_t:/ # echo 0 > /proc/xenomai/latency
rk3568_t:/ # /usr/demo/cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 7.65 7.80 7.75 8/1118 3998

T: 0 ( 3492) P:99 I:1000 C:21069118 Min:      2 Act:    4 Avg:   13 Max:      85
T: 1 ( 3493) P:99 I:1500 C:14046070 Min:      2 Act:   13 Avg:   15 Max:      76
T: 2 ( 3494) P:99 I:2000 C:10534550 Min:      2 Act:    6 Avg:   16 Max:      70
T: 3 ( 3495) P:99 I:2500 C:8427637 Min:      2 Act:   15 Avg:   15 Max:      96
(Testing for 5 hours)
```

2.1.3 Pressure Test 2

`stress-ng -c 4 --io 2 --vm 1 --vm-bytes 4M --timeout 1000000s` + Fishing Expert

2.1.3.1 PREEMPT_RT Patch

```
rk3568_t:/ # cyclictest -c 0 -m -a -t 4 -p 99
policy: fifo: loadavg: 11.92 12.59 12.59 9/1196 3372

T: 0 ( 1389) P:99 I:1000 C:252044459 Min:      0 Act:    6 Avg:    4 Max:     101
T: 1 ( 1390) P:99 I:1500 C:168029630 Min:      0 Act:    3 Avg:    4 Max:      93
T: 2 ( 1391) P:99 I:2000 C:126022214 Min:      0 Act:    1 Avg:    4 Max:     121
T: 3 ( 1392) P:99 I:2500 C:100817765 Min:      0 Act:    6 Avg:    4 Max:     104
(Testing for 24 hours)
```

2.1.3.2 Xenomai Cobalt Mode

```
rk3568_t:/ # echo 0 > /proc/xenomai/latency
rk3568_t:/ # /usr/demo/cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 10.21 10.06 9.99 10/1193 3177

T: 0 ( 2062) P:99 I:1000 C:41956558 Min:      2 Act:   34 Avg:   20 Max:     112
T: 1 ( 2063) P:99 I:1500 C:27971036 Min:      2 Act:   29 Avg:   20 Max:     128
T: 2 ( 2064) P:99 I:2000 C:20978270 Min:      2 Act:   30 Avg:   18 Max:      82
T: 3 ( 2065) P:99 I:2500 C:16782616 Min:      2 Act:   28 Avg:   18 Max:     105
# (Test duration: 12 hours)
```

2.2 RK3562

Configuration	Description
CPU	4xA53, 2016MHz
CACHE	L1-32K, L2-512K
DDR	LP4X, 1320M
OS	BUILDROOT, kernel-5.10

2.2.1 No-Load Test

2.2.1.1 PREEMPT_RT Patch

```
root@rk3562-buildroot:/# cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 1.00 1.26 1.31 1/202 1344

T: 0 ( 1152) P:99 I:1000 C:51905271 Min:      4 Act:    6 Avg:    6 Max:      46
T: 1 ( 1153) P:99 I:1500 C:34603508 Min:      4 Act:    6 Avg:    6 Max:      55
T: 2 ( 1154) P:99 I:2000 C:25952627 Min:      4 Act:    6 Avg:    6 Max:      28
T: 3 ( 1155) P:99 I:2500 C:20762098 Min:      4 Act:    6 Avg:    6 Max:      55
(Testing for 14 hours)
```

2.2.1.2 Xenomai Cobalt Mode

```
root@rk3562-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3562-buildroot:/# /usr/demo/cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 0.02 0.02 0.00 1/152 2459

T: 0 ( 2377) P:99 I:1000 C:14773911 Min:      2 Act:    4 Avg:    4 Max:      31
T: 1 ( 2378) P:99 I:1500 C:9849274 Min:      2 Act:    4 Avg:    4 Max:      28
T: 2 ( 2379) P:99 I:2000 C:7386955 Min:      2 Act:    4 Avg:    4 Max:      18
T: 3 ( 2380) P:99 I:2500 C:5909564 Min:      2 Act:    4 Avg:    4 Max:      32
(Testing for 4 hours)
```

2.2.2 Pressure Testing

```
stress -c 4 --io 2 --vm 1 --vm-bytes 256M --timeout 1000000s
```

2.2.2.1 PREEMPT_RT Patch

```
root@rk3562-buildroot:/# cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 9.42 8.33 8.61 9/220 1477

T: 0 ( 1472) P:99 I:1000 C:7971377 Min:      4 Act:    8 Avg:   10 Max:      69
T: 1 ( 1473) P:99 I:1500 C:5314247 Min:      5 Act:   16 Avg:   12 Max:      72
T: 2 ( 1474) P:99 I:2000 C:3985679 Min:      5 Act:   10 Avg:   10 Max:      63
T: 3 ( 1475) P:99 I:2500 C:3188543 Min:      5 Act:   12 Avg:   12 Max:      76
(2-hour test)
```

2.2.2.2 Xenomai Cobalt Mode

```
root@rk3562-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3562-buildroot:/# /usr/demo/cyclictest -c 0 -m -n -t 4 -p 99
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 7.93 7.92 7.92 9/165 2370

T: 0 ( 2266) P:99 I:1000 C:6493284 Min:      2 Act:   11 Avg:    9 Max:      69
T: 1 ( 2267) P:99 I:1500 C:4328849 Min:      2 Act:   12 Avg:   10 Max:      72
T: 2 ( 2268) P:99 I:2000 C:3246637 Min:      2 Act:   11 Avg:    9 Max:      40
T: 3 ( 2269) P:99 I:2500 C:2597305 Min:      2 Act:   13 Avg:    9 Max:      73
(Testing for 2 hours)
```

2.3 RK3588

Configuration	Description
CPU	4xA76(big), 2.2GHz, 4xA55(little), 1.8GHz
CACHE	A76: L1-64K, L2-512K A55: L1-32K, L2-128K
DDR	LP4X, 2112M
OS	BUILDROOT, kernel-5.10

2.3.1 No-load Test

2.3.1.1 PREEMPT_RT Patch

```
root@rk3588:/# cyclictest -c 0 -m -t 8 -p 99
# /dev/cpu_dma_latency set to 0us

policy: fifo: loadavg: 2.71 3.24 3.29 1/322 3955
T: 0 ( 3406) P:99 I:1000 C:6766462 Min:      3 Act:    4 Avg:    4 Max:      11
T: 1 ( 3407) P:99 I:1500 C:4510976 Min:      4 Act:    4 Avg:    4 Max:      13
T: 2 ( 3408) P:99 I:2000 C:3383228 Min:      4 Act:    4 Avg:    4 Max:      13
T: 3 ( 3409) P:99 I:2500 C:2706579 Min:      4 Act:    4 Avg:    4 Max:      13
T: 4 ( 3410) P:99 I:3000 C:2255481 Min:      1 Act:    1 Avg:    1 Max:       8
T: 5 ( 3411) P:99 I:3500 C:1933267 Min:      1 Act:    1 Avg:    1 Max:       7
T: 6 ( 3412) P:99 I:4000 C:1691609 Min:      1 Act:    1 Avg:    1 Max:       8
T: 7 ( 3413) P:99 I:4500 C:1503652 Min:      1 Act:    1 Avg:    1 Max:       8
(2-hour test)
```

2.3.1.2 Xenomai Cobalt Mode

```
root@rk3588:/# echo 0 > /proc/xenomai/latency
root@rk3588:/# /usr/demo/cyclictest -c 0 -m -n -t 8 -p 99
# /dev/cpu_dma_latency set to 0us

policy: fifo: loadavg: 0.63 0.60 0.55 1/206 1336
T: 0 ( 1163) P:99 I:1000 C:43237018 Min:      1 Act:    2 Avg:    2 Max:       6
T: 1 ( 1164) P:99 I:1500 C:28824678 Min:      1 Act:    2 Avg:    2 Max:       6
T: 2 ( 1165) P:99 I:2000 C:21618508 Min:      1 Act:    2 Avg:    2 Max:       6
T: 3 ( 1166) P:99 I:2500 C:17294807 Min:      1 Act:    2 Avg:    2 Max:       6
T: 4 ( 1167) P:99 I:3000 C:14412339 Min:      1 Act:    2 Avg:    2 Max:       6
T: 5 ( 1168) P:99 I:3500 C:12353433 Min:      1 Act:    2 Avg:    2 Max:       6
T: 6 ( 1169) P:99 I:4000 C:10809254 Min:      1 Act:    2 Avg:    2 Max:       6
T: 7 ( 1170) P:99 I:4500 C:9608226 Min:       1 Act:    2 Avg:    2 Max:       6
(Testing for 12 hours)
```

2.3.2 Pressure Testing

```
stress-ng -c 8 --io 2 --vm 1 --vm-bytes 1024M --timeout 1000000s
```

2.3.2.1 PREEMPT_RT Patch

```
root@rk3588:/# cyclictest -c 0 -m -t 8 -p 99
# /dev/cpu_dma_latency set to 0us

policy: fifo: loadavg: 15.20 15.28 15.29 11/345 10678
T: 0 ( 6379) P:99 I:1000 C:6473077 Min:      3 Act:   13 Avg:    7 Max:      28
T: 1 ( 6380) P:99 I:1500 C:4315388 Min:      4 Act:    7 Avg:    7 Max:      24
T: 2 ( 6381) P:99 I:2000 C:3236536 Min:      3 Act:    4 Avg:    6 Max:      24
T: 3 ( 6382) P:99 I:2500 C:2589231 Min:      4 Act:    8 Avg:    7 Max:      30
T: 4 ( 6383) P:99 I:3000 C:2157683 Min:      1 Act:    1 Avg:    2 Max:      16
T: 5 ( 6384) P:99 I:3500 C:1849443 Min:      1 Act:    2 Avg:    2 Max:      15
T: 6 ( 6385) P:99 I:4000 C:1618261 Min:      1 Act:    2 Avg:    2 Max:      16
T: 7 ( 6386) P:99 I:4500 C:1438455 Min:      1 Act:    2 Avg:    2 Max:      15
(2-hour test)
```

2.3.2.2 Xenomai Cobalt Mode

```
root@rk3588:/# echo 0 > /proc/xenomai/latency
root@rk3588:/# /usr/demo/cyclictest -c 0 -m -n -t 8 -p 99
# /dev/cpu_dma_latency set to 0us

policy: fifo: loadavg: 12.82 12.81 12.81 11/225 1490
T: 0 ( 1179) P:99 I:1000 C:12797171 Min:      1 Act:    4 Avg:    4 Max:      19
T: 1 ( 1180) P:99 I:1500 C:8531447 Min:      1 Act:    4 Avg:    3 Max:      16
T: 2 ( 1181) P:99 I:2000 C:6398585 Min:      1 Act:    4 Avg:    4 Max:      25
T: 3 ( 1182) P:99 I:2500 C:5118868 Min:      1 Act:    3 Avg:    5 Max:      28
T: 4 ( 1183) P:99 I:3000 C:4265723 Min:      1 Act:    5 Avg:    4 Max:      17
T: 5 ( 1184) P:99 I:3500 C:3656334 Min:      1 Act:    4 Avg:    5 Max:      18
T: 6 ( 1185) P:99 I:4000 C:3199292 Min:      2 Act:    4 Avg:    5 Max:      17
T: 7 ( 1186) P:99 I:4500 C:2843815 Min:      1 Act:    5 Avg:    5 Max:      19
(Testing for 4 hours)
```

2.4 RK3576

Configuration	Description
CPU	4xA72(big): 2.3GHz, 4xA53(little): 2.2GHz
CACHE	A72: L1-I-48K, L1-D-32K, L2-1M A53: L1-I-32K, L1-D-32K, L2-512K
DDR	LP4X, 2112M
OS	BUILDROOT, kernel-6.1

2.4.1 No-Load Test

2.4.1.1 PREEMPT_RT Patch

```
root@rk3576-buildroot:/# cyclictest -c 0 -m -t 8 -a 0-7 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 3.68 4.29 4.45 1/328 9247

T: 0 ( 6836) P:99 I:1000 C:86399993 Min:      3 Act:    4 Avg:    3 Max:      29
T: 1 ( 6837) P:99 I:1500 C:57599994 Min:      3 Act:    7 Avg:    5 Max:      33
T: 2 ( 6838) P:99 I:2000 C:43199993 Min:      3 Act:    3 Avg:    3 Max:      31
T: 3 ( 6839) P:99 I:2500 C:34559993 Min:      3 Act:    3 Avg:    3 Max:      29
T: 4 ( 6840) P:99 I:3000 C:28799993 Min:      2 Act:    3 Avg:    3 Max:      16
T: 5 ( 6841) P:99 I:3500 C:24685707 Min:      2 Act:    2 Avg:    3 Max:      20
T: 6 ( 6842) P:99 I:4000 C:21599993 Min:      2 Act:    3 Avg:    3 Max:      13
T: 7 ( 6843) P:99 I:4500 C:19199993 Min:      2 Act:    3 Avg:    3 Max:      20
(Testing for 24 hours)
```

2.4.1.2 Xenomai Cobalt Mode

```
root@rk3576-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3576-buildroot:/# /usr/demo/cyclictest -c 0 -m -t 8 -a 0-7 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 0.30 0.18 0.06 1/227 4479

T: 0 ( 2043) P:99 I:1000 C:86399996 Min:      2 Act:    2 Avg:    2 Max:      30
T: 1 ( 2044) P:99 I:1500 C:57599997 Min:      2 Act:    2 Avg:    2 Max:      26
T: 2 ( 2045) P:99 I:2000 C:43199998 Min:      1 Act:    3 Avg:    2 Max:      24
T: 3 ( 2046) P:99 I:2500 C:34559998 Min:      2 Act:    2 Avg:    2 Max:      26
T: 4 ( 2047) P:99 I:3000 C:28799998 Min:      1 Act:    3 Avg:    2 Max:      17
T: 5 ( 2048) P:99 I:3500 C:24685713 Min:      1 Act:    3 Avg:    2 Max:      16
T: 6 ( 2049) P:99 I:4000 C:21599998 Min:      1 Act:    2 Avg:    2 Max:      16
T: 7 ( 2050) P:99 I:4500 C:19199998 Min:      1 Act:    2 Avg:    2 Max:      17
(Testing for 24 hours)
```

2.4.2 Pressure Test

```
stress-ng -c8 --io 8 --cpu-load 100 --vm 4 --vm-bytes 512M --timeout 10000000s
```

2.4.2.1 PREEMPT_RT Patch

```
root@rk3576-buildroot:/# cyclictest -c 0 -m -t 8 -a 0-7 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us

policy: fifo: loadavg: 16.30 16.36 16.40 13/344 5989
T: 0 ( 4420) P:99 I:1000 C:86399995 Min:      3 Act:    5 Avg:    6 Max:      44
T: 1 ( 4421) P:99 I:1500 C:57599995 Min:      3 Act:    6 Avg:    8 Max:      57
T: 2 ( 4422) P:99 I:2000 C:43199995 Min:      3 Act:    6 Avg:    7 Max:      40
T: 3 ( 4423) P:99 I:2500 C:34559993 Min:      3 Act:    8 Avg:    8 Max:      46
T: 4 ( 4424) P:99 I:3000 C:28799993 Min:      2 Act:    3 Avg:    5 Max:      32
T: 5 ( 4425) P:99 I:3500 C:24685705 Min:      2 Act:    5 Avg:    5 Max:      27
T: 6 ( 4426) P:99 I:4000 C:21599992 Min:      2 Act:    4 Avg:    5 Max:      29
T: 7 ( 4427) P:99 I:4500 C:19199992 Min:      2 Act:    7 Avg:    6 Max:      30
(Testing for 24 hours)
```

2.4.2.2 Xenomai Cobalt Mode

```
root@rk3576-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3576-buildroot:/# /usr/demo/cyclictest -c 0 -m -t 8 -a 0-7 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 12.70 12.57 12.51 13/244 6744

T: 0 ( 4356) P:99 I:1000 C:86399994 Min:      2 Act:    6 Avg:    6 Max:      41
T: 1 ( 4357) P:99 I:1500 C:57599998 Min:      2 Act:   14 Avg:    9 Max:      50
T: 2 ( 4358) P:99 I:2000 C:43199997 Min:      2 Act:    5 Avg:    7 Max:      37
T: 3 ( 4359) P:99 I:2500 C:34559997 Min:      2 Act:    8 Avg:    8 Max:      46
T: 4 ( 4360) P:99 I:3000 C:28799998 Min:      2 Act:    3 Avg:    6 Max:      30
T: 5 ( 4361) P:99 I:3500 C:24685712 Min:      2 Act:    4 Avg:    6 Max:      34
T: 6 ( 4362) P:99 I:4000 C:21599998 Min:      2 Act:    4 Avg:    7 Max:      34
T: 7 ( 4363) P:99 I:4500 C:19199998 Min:      2 Act:    3 Avg:    6 Max:      26
(Testing for 24 hours)
```

2.5 RK3506

Configuration	Description
CPU	3xA7,1512MHz
CACHE	L1-16K,L2-128K
DDR	DDR3,800M
OS	BUILDROOT,kernel-6.1

2.5.1 No-Load Test

2.5.1.1 PREEMPT_RT Patch

```
root@rk3506-buildroot:/# cyclictest -m -a -p 99 -t 3 -i 1000 -D 2h
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 1.72 1.60 1.48 2/127 4000

T: 0 (  502) P:99 I:1000 C:7199993 Min:      0 Act:    0 Avg:    1 Max:      14
T: 1 (  503) P:99 I:1500 C:4799988 Min:      0 Act:    1 Avg:    1 Max:      19
T: 2 (  504) P:99 I:2000 C:3599985 Min:      0 Act:    2 Avg:    1 Max:      26
(Testing for 2 hours)
```

2.5.1.2 Xenomai Cobalt Mode

```
root@rk3506-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3506-buildroot:/# /usr/demo/cyclictest -m -a -p 99 -t 3 -i 1000 -D 2h
WARN: stat /dev/cpu_dma_latency failed: No such file or directory
policy: fifo: loadavg: 0.56 0.31 0.21 1/95 2088

T: 0 ( 2356) P:99 I:1000 C:7199998 Min:      4 Act:    4 Avg:    7 Max:      46
T: 1 ( 2357) P:99 I:1500 C:4799998 Min:      4 Act:   12 Avg:    7 Max:      46
T: 2 ( 2358) P:99 I:2000 C:3599997 Min:      3 Act:    5 Avg:    9 Max:      55
(Testing for 2 hours)
```

2.5.2 Pressure Test

`stress-ng -c 3 --io 2 --vm 1 --vm-bytes 4M --timeout 1000000s`

2.5.2.1 PREEMPT_RT Patch

```
root@rk3506-buildroot:/# cyclictest -m -a -p 99 -t 3 -i 1000 -D 2h
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 8.49 7.88 7.76 7/137 1083

T: 0 ( 1537) P:99 I:1000 C:7199996 Min:      0 Act:    5 Avg:    4 Max:      71
T: 1 ( 1538) P:99 I:1500 C:4799976 Min:      0 Act:    5 Avg:    4 Max:     115
T: 2 ( 1540) P:99 I:2000 C:3599957 Min:      0 Act:    4 Avg:    4 Max:     129
(Testing for 2 hours)
```

2.5.2.2 Xenomai Cobalt Mode

```
root@rk3506-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3506-buildroot:/# /usr/demo/cyclictest -m -a -p 99 -t 3 -i 1000 -D 2h
WARN: stat /dev/cpu_dma_latency failed: No such file or directory
policy: fifo: loadavg: 6.87 6.82 6.72 7/104 1531

T: 0 ( 1919) P:99 I:1000 C:7199991 Min:      4 Act:   20 Avg:   26 Max:     124
T: 1 ( 1920) P:99 I:1500 C:4799993 Min:      4 Act:   21 Avg:   28 Max:     155
T: 2 ( 1921) P:99 I:2000 C:3599987 Min:      5 Act:   16 Avg:   28 Max:     133
(Testing for 2 hours)
```

2.5.3 Isolate CPU Test

Isolate CPU1 and add pressure on the other CPUs:

```
stress-ng -c 3 --io 2 --vm 1 --vm-bytes 4M --timeout 1000000s
```

2.5.3.1 PREEMPT_RT Patch

```
root@rk3506-buildroot:/# cyclictest -m -a 1 -p 99 -t 1 -i 1000 --mainaffinity=0 -D 2h
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 7.19 7.17 7.22 7/137 648

T: 0 ( 1117) P:99 I:1000 C:7199996 Min:      0 Act:    2 Avg:    3 Max:      62
(Testing for 2 hours)
```

2.5.3.2 Xenomai Cobalt Mode

```
root@rk3506-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3506-buildroot:/# /usr/demo/cyclictest -m -a 1 -p 99 -t 1 -i 1000 -D 2h
WARN: stat /dev/cpu_dma_latency failed: No such file or directory
policy: fifo: loadavg: 6.53 6.52 6.57 8/101 1115

T: 0 ( 1503) P:99 I:1000 C:7199992 Min:      4 Act:   15 Avg:   16 Max:      68
(Testing for 2 hours)
```

2.6 RV1126B

Configuration	Description
CPU	4xA53, 1608MHz
CACHE	L1-32K, L2-512K
DDR	DDR3, 1056M
OS	BUILDROOT, kernel-6.1

2.6.1 No-Load Test

2.6.1.1 PREEMPT_RT Patch

```
root@rv1126b-buildroot:/# cyclictest -c 0 -m -a -t 4 -p 99 -D 12h

policy: fifo: loadavg: 1.48 1.70 1.85 1/216 2449

T: 0 ( 1907) P:99 I:1000 C:43199996 Min:      0 Act:    0 Avg:    0 Max:      13
T: 1 ( 1908) P:99 I:1500 C:28799994 Min:      0 Act:    0 Avg:    0 Max:      11
T: 2 ( 1909) P:99 I:2000 C:21599993 Min:      0 Act:    0 Avg:    0 Max:      11
T: 3 ( 1910) P:99 I:2500 C:17279992 Min:      0 Act:    0 Avg:    0 Max:      14
(Testing for 12 hours)
```

2.6.1.2 Xenomai Cobalt Mode

```
root@rv1126b-buildroot:/# echo 0 > /proc/xenomai/latency
root@rv1126b-buildroot:/# /usr/demo/cyclictest -m -n -c 0 -p 99 -t -D 12h
policy: fifo: loadavg: 0.04 0.03 0.00 1/151 2395

T: 0 ( 1827) P:99 I:1000 C:43199992 Min:      2 Act:    2 Avg:    2 Max:      37
T: 1 ( 1828) P:99 I:1500 C:28799995 Min:      2 Act:    4 Avg:    3 Max:      42
T: 2 ( 1829) P:99 I:2000 C:21599996 Min:      2 Act:    2 Avg:    3 Max:      33
T: 3 ( 1830) P:99 I:2500 C:17279996 Min:      2 Act:    3 Avg:    3 Max:      35
(Testing for 12 hours)
```

2.6.2 Pressure Testing

`stress-ng -c 4 --io 2 --vm 2 --vm-bytes 256M --timeout 1000000s`

2.6.2.1 PREEMPT_RT Patch

```
root@rv1126b-buildroot:/# cyclictest -c 0 -m -a -t 4 -p 99 -D 24h

policy: fifo: loadavg: 10.80 11.74 12.03 11/232 3403

T: 0 ( 1915) P:99 I:1000 C:82647713 Min:      0 Act:    3 Avg:    2 Max:      38
T: 1 ( 1916) P:99 I:1000 C:82647712 Min:      0 Act:    2 Avg:    2 Max:      43
T: 2 ( 1917) P:99 I:1000 C:82647709 Min:      0 Act:    3 Avg:    2 Max:      47
T: 3 ( 1918) P:99 I:1000 C:82647698 Min:      0 Act:    3 Avg:    3 Max:      58
(Testing for 24 hours)
```

2.6.2.2 Xenomai Cobalt Mode

```
root@rv1126b-buildroot:/# echo 0 > /proc/xenomai/latency
root@rv1126b-buildroot:/# /usr/demo/cyclictest -m -n -c 0 -p 99 -t -D 24h

policy: fifo: loadavg: 9.06 8.88 8.84 9/166 3460

T: 0 ( 1837) P:99 I:1000 C:86399995 Min:      2 Act:    6 Avg:    7 Max:      91
T: 1 ( 1838) P:99 I:1000 C:86399979 Min:      2 Act:    8 Avg:    8 Max:      70
T: 2 ( 1839) P:99 I:1000 C:86399963 Min:      2 Act:    8 Avg:    6 Max:      41
T: 3 ( 1840) P:99 I:1000 C:86399939 Min:      2 Act:    8 Avg:    9 Max:      72
(Testing for 24 hours)
```

2.7 RK3572

Configuration	Description
CPU	6xA53(little):2.1GHz,2xA73(big):2.2GHz
CACHE	Cluster0(4xA53):L1-I-32K,L1-D-32K,L2-512K Cluster1(2xA53):L1-I-32K,L1-D-32K,L2-256K Cluster2(2xA73):L1-I-64K,L1-D-32K,L2-512K
DDR	LP4X,2112M
OS	BUILDROOT,kernel-6.12

Optimization Notes:

1. The ddr.bin for RK3572 has been specifically optimized for real-time systems. In the U-Boot project, you need to use ./make.sh rk3572-rt to generate loader.bin, which packages the DDR initialization firmware with real-time configuration integrated.
2. The RK3572 is divided into 3 clusters, with each cluster's L2 cache being independent of one another. It is recommended to isolate CPU4-5 (Cluster1) or CPU6-7 (Cluster2) as real-time cores. The isolated CPUs will have higher priority when accessing DDR.

2.7.1 No-Load Test

2.7.1.1 PREEMPT_RT Patch

```
root@rk3572-buildroot:/# cyclictest -c 0 -m -t 8 -a 0-7 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 0.11 0.33 0.52 1/307 5028

T: 0 ( 2502) P:99 I:1000 C:86399990 Min:      3 Act:    3 Avg:    3 Max:      34
T: 1 ( 2503) P:99 I:1500 C:57599990 Min:      3 Act:    4 Avg:    3 Max:      34
T: 2 ( 2504) P:99 I:2000 C:43199990 Min:      3 Act:    4 Avg:    3 Max:      23
T: 3 ( 2505) P:99 I:2500 C:34559991 Min:      3 Act:    3 Avg:    3 Max:      27
T: 4 ( 2506) P:99 I:3000 C:28799991 Min:      3 Act:    4 Avg:    4 Max:      31
T: 5 ( 2507) P:99 I:3500 C:24685705 Min:      3 Act:    4 Avg:    4 Max:      32
T: 6 ( 2508) P:99 I:4000 C:21599991 Min:      3 Act:    4 Avg:    3 Max:      35
T: 7 ( 2509) P:99 I:4500 C:19199991 Min:      3 Act:    3 Avg:    3 Max:      20
(Testing for 24 hours)
```

2.7.1.2 Xenomai Cobalt Mode

```
root@rk3572-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3572-buildroot:/# /usr/demo/cyclictest -c 0 -m -t 8 -a 0-7 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 0.03 0.04 0.01 1/233 11247

T: 0 ( 7387) P:99 I:1000 C:86399996 Min:      2 Act:    2 Avg:    2 Max:      26
T: 1 ( 7388) P:99 I:1500 C:57599997 Min:      2 Act:    2 Avg:    2 Max:      25
T: 2 ( 7389) P:99 I:2000 C:43199997 Min:      2 Act:    2 Avg:    3 Max:      29
T: 3 ( 7390) P:99 I:2500 C:34559998 Min:      2 Act:    3 Avg:    3 Max:      22
T: 4 ( 7391) P:99 I:3000 C:28799998 Min:      2 Act:    3 Avg:    3 Max:      22
T: 5 ( 7392) P:99 I:3500 C:24685712 Min:      2 Act:    3 Avg:    3 Max:      20
T: 6 ( 7393) P:99 I:4000 C:21599998 Min:      2 Act:    3 Avg:    3 Max:      19
T: 7 ( 7394) P:99 I:4500 C:19199998 Min:      2 Act:    3 Avg:    3 Max:      17
(Testing for 24 hours)
```

2.7.2 Pressure Testing

```
stress-ng -c 8 --io 2 --vm 2 --vm-bytes 512M --timeout 1000000s
```

2.7.2.1 PREEMPT_RT Patch

```
root@rk3572-buildroot:/# cyclictest -c 0 -m -t 8 -a 0-7 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 13.67 13.67 13.60 14/325 5269

T: 0 ( 2552) P:99 I:1000 C:86399989 Min:      3 Act:    4 Avg:    6 Max:     103
T: 1 ( 2553) P:99 I:1500 C:57599990 Min:      3 Act:    7 Avg:    9 Max:     114
T: 2 ( 2554) P:99 I:2000 C:43199988 Min:      3 Act:    6 Avg:    8 Max:     108
```

```
T: 3 ( 2555) P:99 I:2500 C:34559987 Min:      3 Act:    6 Avg:    8 Max:     126
T: 4 ( 2556) P:99 I:3000 C:28799987 Min:      3 Act:    9 Avg:   11 Max:      79
T: 5 ( 2557) P:99 I:3500 C:24685700 Min:      3 Act:    5 Avg:   11 Max:      88
T: 6 ( 2558) P:99 I:4000 C:21599985 Min:      4 Act:   10 Avg:   11 Max:      98
T: 7 ( 2559) P:99 I:4500 C:19199984 Min:      4 Act:   12 Avg:   10 Max:     105
(Testing for 24 hours)

Isolate CPU4~5 (add "isolcpus=4-5 nohz_full=4-5" to kernel cmdline):
root@rk3572-buildroot:/# cyclictest -c 0 -m -t 2 -a 4-5 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 14.13 14.17 14.08 16/315 6997

T: 0 ( 3563) P:99 I:1000 C:86399998 Min:      3 Act:    3 Avg:    3 Max:      27
T: 1 ( 3564) P:99 I:1500 C:57599995 Min:      3 Act:    3 Avg:    3 Max:      19
(Testing for 24 hours)
```

2.7.2.2 Xenomai Cobalt Mode

```
root@rk3572-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3572-buildroot:/# /usr/demo/cyclictest -c 0 -m -t 8 -a 0-7 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 13.09 13.12 13.09 13/261 7378

T: 0 ( 2547) P:99 I:1000 C:86399996 Min:      2 Act:    6 Avg:    6 Max:     107
T: 1 ( 2548) P:99 I:1500 C:57599997 Min:      2 Act:    5 Avg:    8 Max:     109
T: 2 ( 2549) P:99 I:2000 C:43199997 Min:      2 Act:    4 Avg:    9 Max:     123
T: 3 ( 2550) P:99 I:2500 C:34559996 Min:      2 Act:    4 Avg:    9 Max:     119
T: 4 ( 2551) P:99 I:3000 C:28799997 Min:      2 Act:    9 Avg:   14 Max:     113
T: 5 ( 2552) P:99 I:3500 C:24685711 Min:      3 Act:   11 Avg:   13 Max:     105
T: 6 ( 2553) P:99 I:4000 C:21599996 Min:      3 Act:   10 Avg:    9 Max:      91
T: 7 ( 2554) P:99 I:4500 C:19199996 Min:      4 Act:   13 Avg:   11 Max:     110
(Testing for 24 hours)

Isolate CPU4~5 (add "isolcpus=4-5 nohz_full=4-5" to kernel cmdline):
root@rk3572-buildroot:/# echo 0 > /proc/xenomai/latency
root@rk3572-buildroot:/# /usr/demo/cyclictest -c 0 -m -t 2 -a 4-5 -p 99 -D24H
# /dev/cpu_dma_latency set to 0us
policy: fifo: loadavg: 13.38 13.47 13.52 12/231 5898

T: 0 ( 2444) P:99 I:1000 C:86399996 Min:      2 Act:    3 Avg:    3 Max:      34
T: 1 ( 2445) P:99 I:1500 C:57599997 Min:      2 Act:    3 Avg:    3 Max:      30
(Testing for 24 hours)
```